

A Use Case on a Multi-Energy System related to PEPR Future Energy Networks (Réseaux du Futur)

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Physical Twin description

- Experimental **multi-energy system & network**
- **Micro-grid** including solar PV, batteries, gensets, ...
- **District heating and cooling network** including heat pump, solar thermal, thermal storage, ...

Digital Twin usages

- To **predict** solar production + heating, cooling & power demand
- To **estimate system state** from partial measurements (esp. thermal storage)
- To **compute optimal control** setpoints for each equipment
- To **detect and correct errors & faults**, in order to maintain the most operation efficient & robust

Digital Twin Description

- Physical and ML **models of physical components**, power and thermal networks, demand
- Model **synchronization from measurements**
- **Set-points** based on modeling and optimization
- **Multi-level** control architecture (on going)

Engineering Digital Twin challenges

- **WP4.1** Implementing smart data collection strategies.
- **WP4.2** Synchronizing models with real-world data.
- **WP4.3** Adapting/calibrating DT using real-world data.
- **WP4.4** Optimizing networks for digital coupling.
- **WP2.1** Implementing a reference architecture for managing variability, fidelity and uncertainty.
- **WP2.2** Modularizing for composability (intra & inter DT).
- **WP1.1** Creating interfaces for system hybridization.
- **WP3.1** Analyzing hybrid models composition.

One multi-energy, multi-scale DT for the whole energy system ? (optimal dispatch, market participation,...)

DTs for individual systems and networks ? (prediction, optimal control, fault diagnosis ...)

DTs for individual equipment ? (control, estimation, fault detection, predictive maintenance ...)

